A Surrogate Model for Reducing the Computational Cost of Neuron Simulations

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Introduction

- Multi-compartment models represent the spatial morphology of a neuron as connected compartments, reproducing neuronal activity in fine detail.
- Each compartment is described by differential equations carrying **gate variables** (ion-channel open probabilities), as in the Hodgkin–Huxley (HH) model.

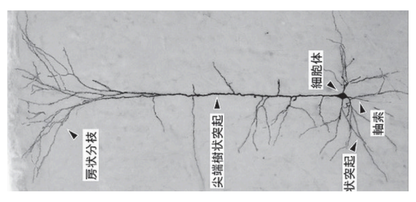
$$C_m \frac{dV}{dt} = -\bar{g}_L(V - E_L) - \bar{g}_{\mathrm{Na}} m^3 h (V - E_{\mathrm{Na}}) - \bar{g}_{\mathrm{K}} n^4 (V - E_{\mathrm{K}}) + I_{\mathrm{ext}}$$

HH membrane-potential equation (m, h, n: gate variables)

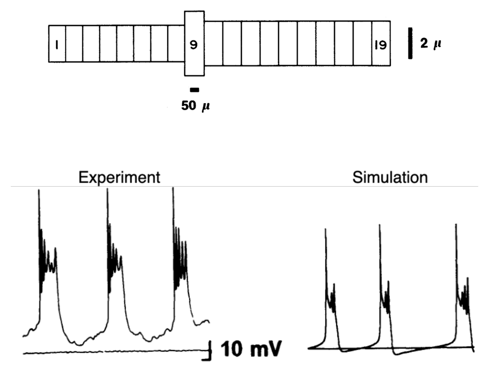
→ When many multi-compartment models are simulated in parallel (e.g. brain-scale simulation), the gate variables cause a **memory bottleneck**.

Goal

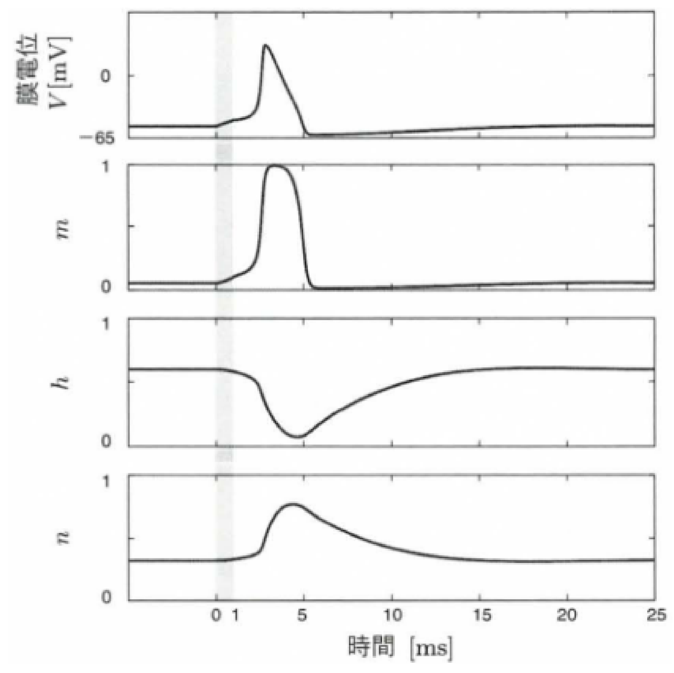
Build a surrogate model that reproduces the membrane-potential response of a multi-compartment neuron with **fewer state variables**.



Pyramidal neuron [1]



Spatial representation and simulation of a multi-compartment model [2]



HH membrane-potential response [3]

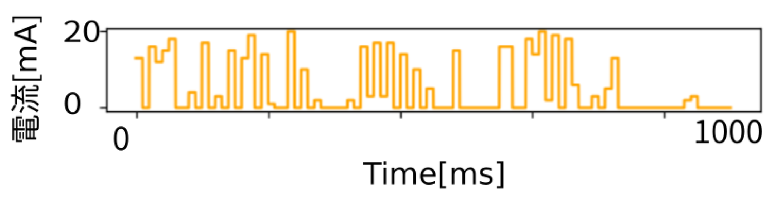
Methods

Two-axis surrogate framework

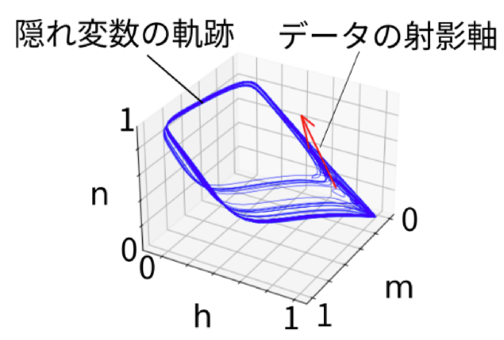
1. Simulate the target model; collect time series of membrane potential V and gate variables.
2. **Compress** the gates into a low-dim latent $z \in \mathbb{R}^n$ with a preprocessor — PCA (linear) or an **autoencoder** (nonlinear).
3. **Identify** the latent's governing equation. The ansatz sets the **assumed form**: **SINDy** picks terms from a generic candidate library, while **hybrid** keeps the known ionic physics and lets SINDy fit **only** the latent.

$$\begin{cases} \frac{dV}{dt} = \underbrace{I_{\mathrm{ion}}^{\mathrm{phys}}(V, \boldsymbol{z})}_{\text{known physics}} + I_{\mathrm{ext}} \\ \frac{d\boldsymbol{z}}{dt} = \underbrace{\Theta(\boldsymbol{z}, V)\boldsymbol{\xi}}_{\text{SINDy-identified}} \\ \boldsymbol{z} = \mathrm{AE}_{\mathrm{enc}}(\text{gates}), \quad \text{gates} = \mathrm{AE}_{\mathrm{dec}}(\boldsymbol{z}) \end{cases}$$

The hybrid ansatz directly fixes last year's **over-large (41-term)** SINDy library, which contained many non-physical terms.



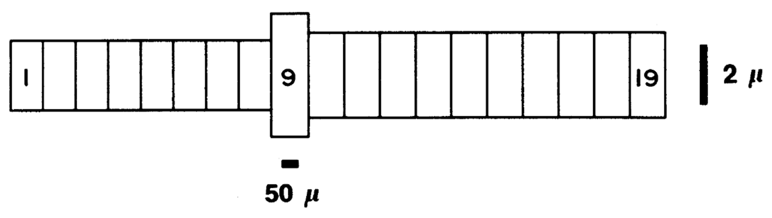
Random 10 ms pulse currents used to drive the simulation



Gate compression into latent space

Target: Traub CA3 pyramidal cell

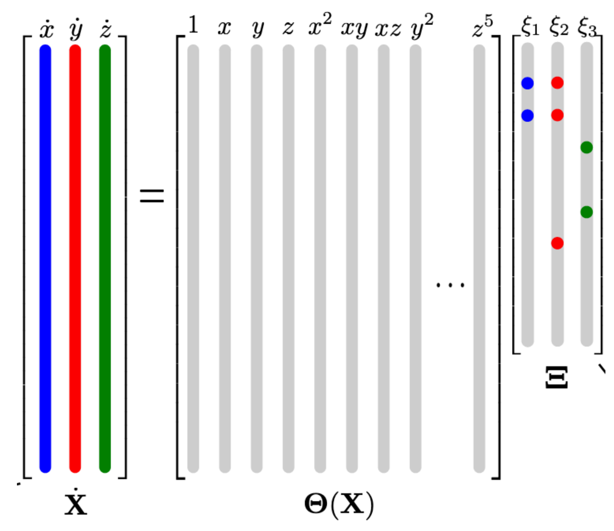
(a 19-compartment model — a realistic multi-compartment neuron, replacing last year's toy 3-compartment model)



We replace the soma compartment with the surrogate, so that the ionic gate variables of the soma are represented by the low-dimensional latent $\boldsymbol{z}$ while the compartment coupling stays physical.

- preprocessor $\in$ {PCA, AE}
- ansatz $\in$ {SINDy, hybrid}
- latent dim $n = 2$ (HH) / 3 (Traub)

SINDy



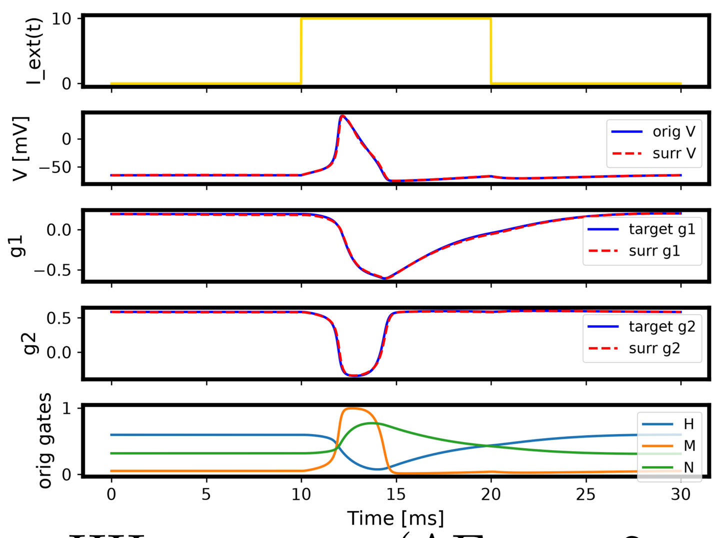
X: time-series data

$\Theta(X)$: library evaluated at X

Solve $\dot{X} = \Theta(X)\Xi$ as a sparse regression for the coefficient matrix Ξ [4]

Results and Discussion

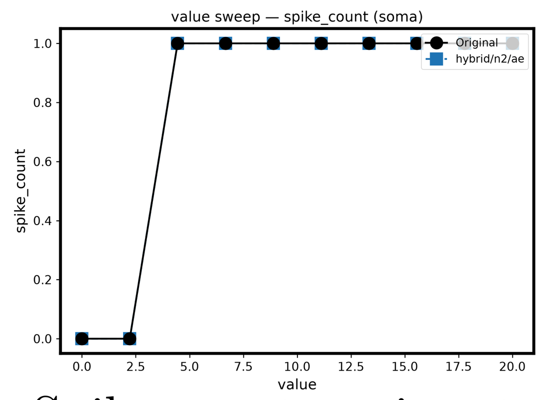
① Proof of concept — single HH soma



HH surrogate (AE, $n = 2$, hybrid): V, the two AE latents g_1, g_2, and the original 3 gates. Original (blue) vs. surrogate (red, dashed) overlap almost exactly.

Waveform match

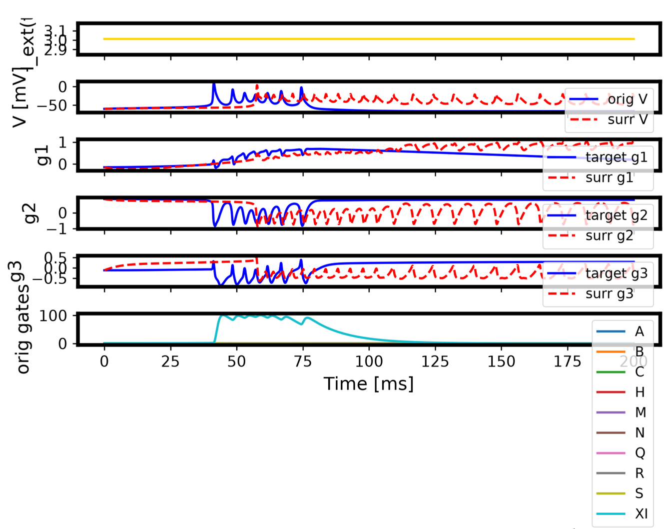
- spike-shape corr = **1.000**
- AP amplitude err = **0.5 mV**
- RMSE = **1.0 mV**



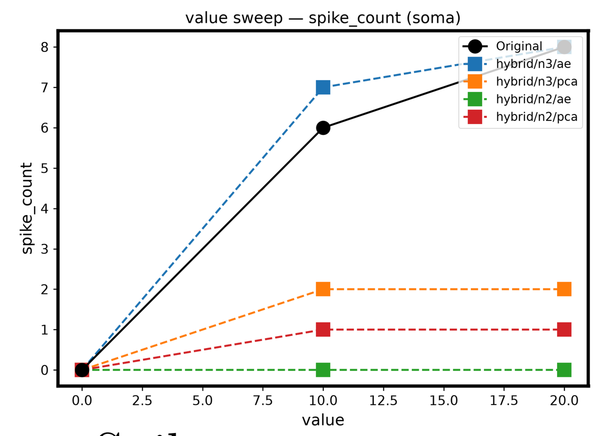
Spike count vs. input amplitude — matches the original over the whole range (not overfit to one stimulus).

- **3 gates $(m, h, n) \to$ 2 latents**: the AE compresses the gates into a 2-D latent, yet the surrogate reproduces the spike with **near-zero error**.
- The hybrid ansatz retains the ionic physics, so the identified latent dynamics stay compact and stable.

② Scaling up — Traub 19-compartment (soma replaced)



Traub19 soma replaced (AE, $n = 3$, hybrid). V, the 3 latents, and the original soma gates. Original (blue) vs. surrogate (red, dashed).



Spike count across surrogate variants. Only **AE + $n = 3$** tracks the original; PCA and $n = 2$ collapse — motivating the nonlinear AE and the $n = 3$ latent.

- The **same AE + hybrid framework** transfers to a realistic multi-compartment model **without diverging**, firing in bursts and tracking the input-driven **spike count** (spike-shape corr = 1.000).
- It captures the **qualitative structure** — firing threshold and burst onset shift with input amplitude, as in the original.
- **Remaining gap**: the post-burst **decay/ termination** is not yet reproduced — the surrogate keeps oscillating.

Conclusion

- For a single HH neuron, the **AE + hybrid** surrogate reproduces the spike waveform with **near-zero error** while halving+ the state dimension ($4 \to 2$).
- The same framework **scales to a realistic 19-compartment Traub model** without diverging and captures the qualitative firing behaviour, though not yet quantitatively.

→ Future work

Reproduce the post-burst decay dynamics of the multi-compartment surrogate, and further refine the physics-informed library.

Code link

Code for this study is available at https://github.com/MunechikaHaruki/SINDyNeuroSurrogate

Bibliography

[1] "錐体細胞." Accessed: Dec. 08, 2025. [Online]. Available: https://doi.org/10.14931/bsd.2091

[2] R. D. Traub, R. K. Wong, R. Miles, and H. Michelson, "A Model of a CA3 Hippocampal Pyramidal Neuron Incorporating Voltage-Clamp Data on Intrinsic Conductances," *Journal of Neurophysiology*, vol. 66, no. 2, pp. 635–650, Aug. 1991, doi: 10.1152/jn.1991.66.2.635.

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[4] K. Champion, B. Lusch, J. N. Kutz, and S. L. Brunton, "Data-Driven Discovery of Coordinates and Governing Equations," *Proceedings of the National Academy of Sciences*, vol. 116, no. 45, pp. 22445–22451, Nov. 2019, doi: 10.1073/pnas.1906995116.